

Nodal analysis

1. Circuit analysis

1.1 Target: to find unknown parameters
current I , V , P , E

1.2 Available tools/laws: ① Ohm's law, R
② KCL; ③ KVL.

1.3 New approaches

2. Introduction to nodal analysis

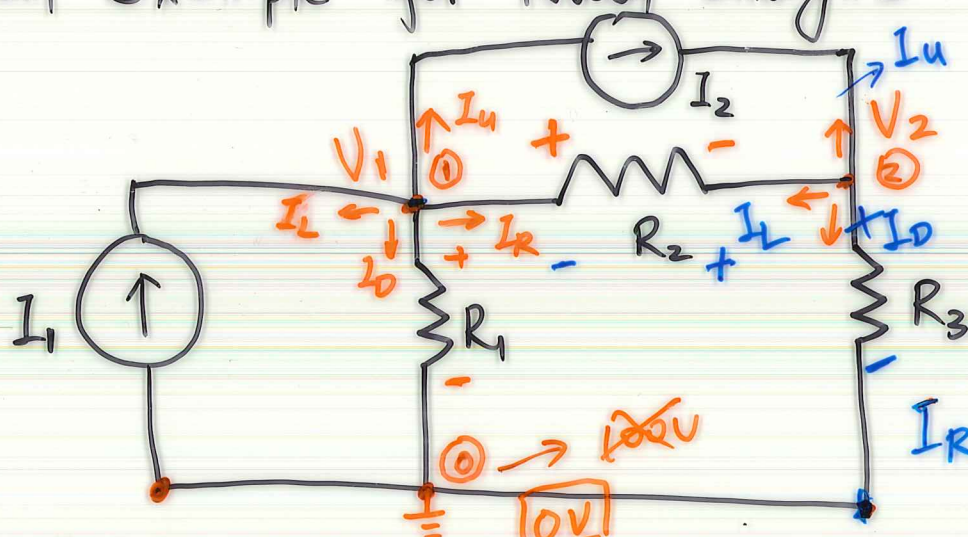
2.1 Concept

- Use KCL to determine node voltages.

2.2 Steps

- node analysis
 - count:
 - select:
 - assign:
- KCL
 - assign current
 - current direction
 - branch current
 - setup equations
- solve equations

Detail example for Nodal analysis



B: 5

N: 3 - 1

2 nodes

$$I_R = \frac{V_{\text{from}} - V_{\text{to}}}{R_2}$$

1. nodes: reference node, critical nodes.

2. KCL equations

3. solve equations

$$I_L = -I_1$$

$$I_R = \frac{V_{\text{high}} - V_{\text{low}}}{R_2} = \frac{V_1 - V_2}{R_2}$$

$$I_u = I_2$$

$$I_D = \frac{V_1 - 0}{R_1} = \frac{V_1}{R_1}$$

$$I_u = -$$

$$I_D = \frac{V_2}{R_3}$$

$$I_L = \frac{V_2 - 0}{R_3}$$

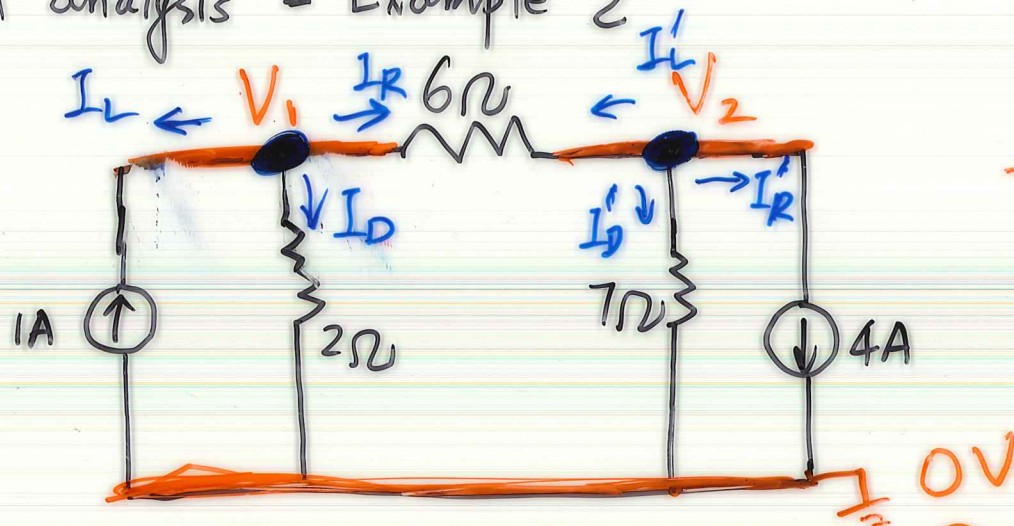
Eg 1: $I_L + I_R + I_u + I_D = 0$

$$-I_1 + \frac{V_1 - V_2}{R_2} + I_2 + \frac{V_1}{R_1} = 0 \quad (1)$$

Eg 2: $I_L + I_u + I_D = 0$

$$\frac{V_2 - V_1}{R_2} + (-I_2) + \frac{V_2}{R_3} = 0 \quad (2)$$

Nodal analysis - Example 2



elimination

Step 1: Branches: 5 ; Nodes: 3,
assign ref node, and unknown voltages

Step 2: $I_L + I_D + I_R = 0$, $-1 + \frac{V_1}{2} + \frac{V_1 - V_2}{6} =$

$$I'_L + I'_D + I_R = 0 \quad \frac{V_2 - V_1}{6} + \frac{V_2}{7} + 4 =$$

$$\textcircled{1} \quad -6 + 3V_1 + V_1 - V_2 = 0$$

$$4V_1 - V_2 = 6 \Rightarrow \underline{V_2 = 4V_1 - 6}$$

$$\textcircled{2} \quad 7V_2 - 7V_1 + 6V_2 + 168 = 0$$

$$13V_2 - 7V_1 + 168 = 0$$

$$13(4V_1 - 6) - 7V_1 = -168$$

$$52V_1 - 7V_1 = -168 + 78 = -90$$

$$45V_1 = -90 \Rightarrow V_1 = -2V$$

$$V_2 = 4V_1 - 6 = -8 - 6 = -14V$$

Nodal analysis - Example 2

Cramer's Rule

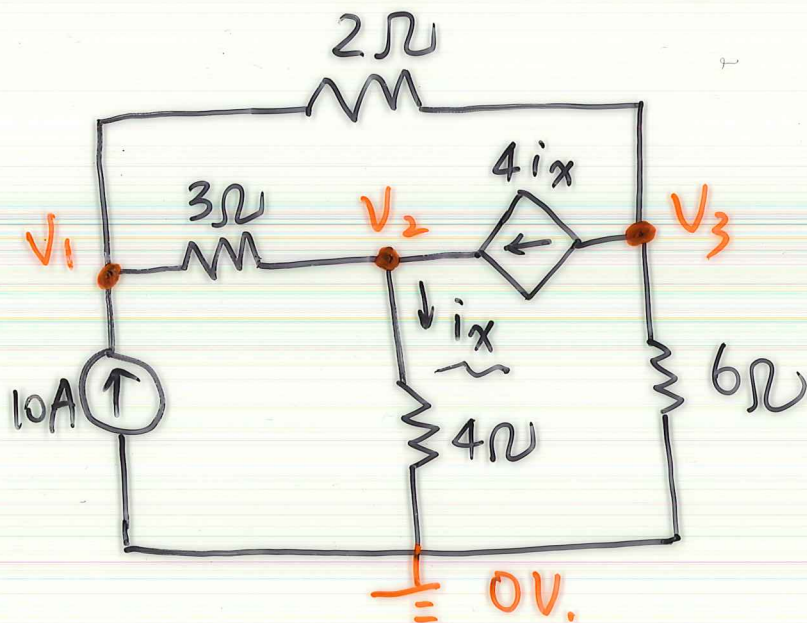
$$\begin{cases} 4V_1 + (-1)V_2 = 6 & (1) \\ (-7)V_1 + 13V_2 = -168 & (2) \end{cases}$$

$$\begin{pmatrix} 4 & -1 \\ -7 & 13 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \end{pmatrix} = \begin{pmatrix} 6 \\ -168 \end{pmatrix}$$

$$V_1 = \frac{\Delta_1}{\Delta} = \frac{\begin{vmatrix} 6 & -1 \\ -168 & 13 \end{vmatrix}}{\begin{vmatrix} 4 & -1 \\ -7 & 13 \end{vmatrix}} = \frac{78 - (-1) \times (-168)}{4 \times 13 - (-1) \times (-7)} = \frac{-90}{52 - 7 = 45} = -2$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{\begin{vmatrix} 4 & 6 \\ -7 & -168 \end{vmatrix}}{\begin{vmatrix} 4 & -1 \\ -7 & 13 \end{vmatrix}} = \frac{4 \times (-168) - 6 \times (-49)}{45} = \frac{-630}{45} = -14$$

Determine nodal voltages.



$$\begin{aligned} \text{CCCS} \\ 4i_x &= \frac{4V_2}{4} \\ i_x &= \frac{V_2}{4} \end{aligned}$$

$$\textcircled{1}: -10 + \frac{V_1 - V_2}{3} + \frac{V_1 - V_3}{2} = 0$$

$$\underline{4i_x = V_2}$$

$$\textcircled{2}: \frac{V_2 - V_1}{3} + (-4i_x) + \frac{V_2}{4} = 0$$

$$\textcircled{3}: \frac{4i_x}{V_2} + \frac{V_3 - V_1}{2} + \frac{V_3}{6} = 0$$

$$\begin{aligned} \textcircled{1} \times 6: -60 + 2V_1 - 2V_2 + 3V_1 - 3V_3 &= 0 \\ 5V_1 - 2V_2 - 3V_3 &= 60 \end{aligned}$$

$$\textcircled{2} \times 12: -4V_1 - 5V_2 + 0V_3 = 0$$

$$\textcircled{3} \times 6: -3V_1 + 6V_2 + 4V_3 = 0$$

$$\begin{pmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{pmatrix} \begin{pmatrix} V_1 \\ V_2 \\ V_3 \end{pmatrix} = \begin{pmatrix} 60 \\ 0 \\ 0 \end{pmatrix} \quad \checkmark$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{\begin{vmatrix} 5 & 60 & -3 \\ -4 & 0 & 0 \\ -3 & 0 & 4 \end{vmatrix}}{\begin{vmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{vmatrix}}$$

$$\begin{vmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -3 & 6 & 4 \end{vmatrix} = -15$$

$$\Delta_2 = \begin{vmatrix} 5 & 60 & -3 \\ -4 & 0 & 0 \\ -3 & 0 & 4 \end{vmatrix}$$

Diagram showing the expansion of the determinant Δ_2 using the first column. The signs for the cofactors are indicated by a checkerboard pattern: +, -, + for the first column. The terms are: $5 \cdot 0 \cdot 4$ (positive), $-(-4) \cdot 0 \cdot (-3)$ (negative), and $-(-3) \cdot 60 \cdot 0$ (negative).

$$= 5 \cdot 0 \cdot 4 + (-4) \cdot 0 \cdot (-3) + (-3) \cdot 60 \cdot 0$$

$$= -(-3) \cdot (0) \cdot (-3) - (0 \cdot 0 \cdot 5) - (-4) \cdot 4 \cdot 60$$

$$= 960,$$

$$\Delta = -15$$